



OCCUPATIONAL CERTIFICATE: Financial Markets Practitioner

Module 2: Fundamentals of Economics STUDY GUIDE

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Learning Outcomes

On completion of this learning unit, you will be able to:

- Distinguish between micro and macro economics
- Explain the concepts GDP, GNP, employment, inflation, exchange rates, interest rates, monetary policy and money supply, fiscal policy and taxes, national budgets, international trade and balance of payments
- Evaluate the influence of political and legislative factors on the macro-economic environment
- Evaluate the effect of macroeconomic occurrences on the financial markets and instruments
- Evaluate the relative impact of a budget deficit and a budget surplus on the financial markets and securities
- Evaluate the relative impact of fiscal and monetary policy on the financial markets and securities
- Explain the concepts of buy-sell decisions, saving and investing, opportunity cost, elasticity, utility, marginal costs and benefits
- Interpret movements along and shifts of the supply and demand curves and explain the reasons and implications
- Explain the influence of political and legislative factors on the micro economic environment
- Apply the law of supply and demand to the financial markets
- Evaluate the rationale for international trade and the concepts of imports and exports
- Analyse and interpret the law of comparative advantage and its effects on international trade
- Interpret balance of payments, including details on current and capital and financial accounts
- Define the field of behavioural economics and explain its relevance and application in finance
- Evaluate the impact and consequences of rational and irrational individual and group behaviour on investments

ECONOMICS

1.1. INTRODUCTION TO ECONOMICS

1.1.1 ECONOMICS: THE STUDY OF CHOICE

Economics is a social science that examines how people choose among the alternatives available to them. It is social because it involves people and their behaviour. It is a science because it uses, as much as possible, a scientific approach in its investigation of choices.

Economics is the study of the allocation of scarce resources and involves making the following decisions:

- What should be produced?
- The manner in which goods and services should be produced (how should they be produced?); and
- For whom should the goods and services be produced?

Economics can be defined as ‘the study of the way in which human beings employ **scarce resources** (with alternative uses) to satisfy their many needs’. Human beings have many needs and wants, but their resources are limited and often insufficient to meet all their needs and wants.



Scarcity is a central concept in economics because it explains the fundamental economic problem of unlimited wants and needs in a world with limited resources. Scarcity can affect a country's money supply, natural resources, available labour, and means of production.

Scarcity may denote a change in a market equilibrium raising the price based on the law of supply and demand. In those instances, scarcity denotes a decrease over time in the supply of the product or commodity relative to demand. The growing scarcity reflected in the higher price required to attain a market equilibrium could be attributable to one or more of the following:

- Demand-induced scarcity reflects rising demand
- Supply-induced scarcity caused by diminished supply
- Structural scarcity attributable to mismanagement or inequality

Scarcity can explain a market shift to a higher price, compare the availability of economic inputs, or convey the opportunity cost in allocating limited resources. The definition of a market price is one at which supply equals demand, meaning all those willing to obtain the resource at a **market price** can do so. Scarcity can explain a market shift to a higher price, compare the availability of economic inputs, or convey the opportunity cost in allocating limited resources

A prominent example of intentional scarcity can be seen in the drug market. Inventors of new drugs and devices secure patents to prevent competitors from manufacturing the same products for a limited amount of time. This intentionally creates scarcity, allowing inventors to benefit commercially from their work for a window.

Monetary Policy Affect Scarcity

When governments print too much money, the value of the money decreases. Supply is high and money is less scarce. However, too much money in an economy can lead to inflation. Governments tend to keep the money supply relatively scarce through contractionary policy. The main contractionary policies employed by the United States include raising interest rates, increasing bank reserve requirements, and selling government securities.

1.1.2 OPPORTUNITY COST

Resources are limited, while needs and wants are unlimited. We are therefore faced with the economic problem of scarcity, which forces us as individuals and as a society to continuously make choices about how our scarce resources are to be used to satisfy our needs and wants. Whenever a choice is made, it involves a cost that is known as the **opportunity cost** of the choice.

Economic opportunity cost is the value of the best alternative that is forgone when a choice is made between several mutually exclusive options. It's also known as alternative cost.

Opportunity cost is a central concept in economics because it helps to understand the true costs of decisions. It's a key factor in decision making because it helps to evaluate the relative advantages and disadvantages of different options

Opportunity cost is used in calculating economic profits, which are theoretical in nature because they account for opportunity costs. Accounting profits are calculated using only explicit costs, so they are higher than economic profits.

- Opportunity cost is the forgone benefit that would have been derived from an option other than the one that was chosen.
- To properly evaluate these costs, the costs and benefits of every option available must be considered and weighed against the others.
- Considering potential opportunity costs can guide individuals and organisations to more profitable decision making.
- This cost of a lost benefit is a strictly internal measure used for strategic planning; it is not included in accounting profit or reflected in external financial reporting.

- Examples of opportunity cost considerations include investing in a new manufacturing plant in Los Angeles as opposed to Mexico City, deciding to upgrade company equipment or hire additional workers, or buying stock A vs. stock B.

We can express opportunity cost in terms of a return (or profit) on investment by using the following mathematical formula. The formula for this calculation is simply the difference between the expected returns of each option:

$$\text{Opportunity Cost} = \text{RMPIC} - \text{RICP}$$

where:

RMPIC = Return on most profitable investment choice

RICP = Return on investment chosen to pursue



CASE STUDY

The Most Expensive Pizza Ever?

One of the most dramatic examples of opportunity cost is a 2010 exchange of 10,000 bitcoins for two large pizzas - at the time worth about \$41.1. As of August 2024, those 10,000 bitcoins would be worth over \$690 million

Opportunity Cost Analysis for an Individual

Individuals also face decisions involving such missed opportunities, even if the stakes are often smaller.

Suppose, for example, that you've just received an unexpected R1,000 bonus at work. You could simply spend it now, such as on a spur-of-the-moment vacation, or invest it for a future trip. For example, if you were to invest the entire amount in a safe, one-year certificate of deposit at 5%, you'd have R1,050 to play with next year at this time.

You'd also face an opportunity cost with your vacation days at work. If you use some of them now with your spare R1,000 you won't have them next year (assuming your employer lets you roll them over from year to year).

As with many similar decisions, there is no right or wrong answer here, but it can be a helpful exercise to think it through and decide what you most want.

Explicit vs Implicit Costs

Company expenses are broadly divided into two categories—explicit costs and implicit costs. The former are expenses like rents, salaries, and other operating expenses that are paid with a company's tangible assets and recorded within a company's financial statements.

By contrast, implicit costs are technically not incurred and cannot be measured accurately for accounting purposes. There are no cash exchanges in the realisation of implicit costs. Instead, they are opportunity costs, making them synonymous with imputed costs, while explicit costs are considered out-of-pocket expenses.

Opportunity Cost vs Risk

In economics, risk describes the possibility that an investment's actual and projected returns will be different and that the investor may lose some or all of their capital. Opportunity cost reflects the possibility that the returns of a chosen investment will be lower than the returns of a forgone investment.

The key difference is that risk compares the actual performance of an investment against the projected performance of the same investment, while opportunity cost compares the projected performance of an investment against the projected performance of another investment.

1.1.3 MARKET PRICE

The market price is the current price at which a product or service can be bought or sold. The market price of a product or service is determined by the forces of supply and demand. The price at which quantity supplied equals quantity demanded is the market price.

The market price is used to calculate consumer and economic surplus.

- Consumer surplus is the difference between the highest price consumers are willing to pay for a product and the actual price they pay, or the market price.
- Economic surplus is comprised of two related quantities: consumer surplus and producer surplus. Producer surplus is profit: It is the amount over cost that a producer obtains by selling at the market price, provided that the market price is higher than the minimum that they would be willing to sell for.

Economic surplus is the sum total of consumer surplus and producer surplus.

- Market price is the current price of a product or service.
- The market price of a product or service is determined by the forces of supply and demand. It's the price at which quantity supplied equals quantity demanded.
- In financial markets, market prices change constantly as people change their bid or offer prices, or as sellers hit the bid or buyers hit the offer.

A shock to either the supply or the demand for a product or service can change the market price for a product or service. A supply shock is an unexpected event that suddenly changes the supply of a good or service. A demand shock is a sudden event that increases or decreases the demand for a good or service.

Some examples of supply shock are interest rate cuts, tax cuts, government stimulus, terrorist attacks, natural disasters, and stock market crashes. Some examples of demand shock include a steep rise in oil and gas prices or other commodities, political turmoil, natural disasters, and breakthroughs in production technology.

1.1.4 MARKET PRICES IN EQUITY

In stock trading, the market price is the most recent price at which a stock was traded.

Stock market prices are the result of the interaction of traders, investors, and dealers. For a trade to occur, a buyer and a seller must agree on a price. Bids are represented by buyers and offers are represented by sellers.

The bid is the highest price someone is advertising they will buy at, while the offer is the lowest price someone else is advertising, they will sell at. For a stock, this may be R50.51 and R50.52.

A buyer who no longer thinks that R50.52 is a good price may drop the bid to R50.25. The sellers may or may not agree. A trade only occurs if a seller accepts the bid price, or a buyer accepts the offer price.

Bids and offers change constantly as buyers and sellers change their bids. As sellers accept bids, the price drops, and as buyers accept offers, the price rises.

In the bond market, the market price in the bond market is the last reported price excluding accrued interest. This is called the clean price.

The market price of a product or service is determined by the law of supply and demand. If the amount supplied is roughly the same as the amount demanded, the price will stay the same. When the amount supplied exceeds the amount demanded, or vice versa, the market price will decline or increase.

1.1.5 MACRO ECONOMICS

Macroeconomics studies the behaviour of a country and how its policies impact the economy as a whole. It analyses entire industries and economies rather than individuals or specific companies. This is why it's referred to as a top-down approach. It tries to answer questions such as "What should the rate of inflation be?" or "What stimulates economic growth?"



Macroeconomics examines economy-wide phenomena such as gross domestic product (GDP) and how GDP is affected by changes in unemployment, national income, rates of growth, and price levels.

Macroeconomics analyses how an increase or decrease in net exports impacts a nation's capital account or how gross domestic product (GDP) is impacted by the unemployment rate.

It focuses on aggregates and econometric correlations. This is why governments, and their agencies rely on macroeconomics to formulate economic and fiscal policy. Investors who buy interest-rate-sensitive securities should keep a close eye on monetary and fiscal policy.

John Maynard Keynes is often credited as the founder of macroeconomics because he initiated the use of monetary aggregates to study broad phenomena. However, some economists dispute his theories, and many Keynesians disagree on how to interpret his work.

Macroeconomic factors can have a significant influence on your investment portfolio. The Great Recession of 2008–09 and the accompanying market crash were caused by the bursting of the U.S. housing bubble and the subsequent near-collapse of financial institutions that were heavily invested in U.S. subprime mortgages.

Consider the response of central banks and governments to the pandemic-induced crash of spring 2020 for another example of the effect of macro factors on investment portfolios. Governments and central banks unleashed torrents of liquidity through fiscal and monetary stimulus to prop up their economies and stave off recession. This pushed most major equity markets to record highs in the second half of 2020 and throughout much of 2021.

1.1.6 MICROECONOMICS

Microeconomics is a logical starting point for the study of economics. Microeconomics is the study of decisions made by people and businesses regarding the allocation of resources and the prices at which they trade goods and services. It considers taxes, regulations, and government legislation

Microeconomics focuses on supply and demand, and other forces that determine price levels in the economy. It takes a bottom-up approach to analysing the economy by attempting to understand human choices, decisions, and the allocation of resources.

Microeconomics doesn't try to answer or explain what forces should take place in a market. Rather, it tries to explain what happens when there are changes in certain conditions.

Microeconomics examines how a company can maximise its production and capacity so it can lower prices and better compete. A lot of microeconomic information can be gleaned from company financial statements. Microeconomics involves several key principles, including but not limited to:

- **Supply, demand, and equilibrium:** Prices are determined by the law of supply and demand. Suppliers offer the same price demanded by consumers in a perfectly competitive market. This creates an economic equilibrium.
- **Production theory:** This is the study of how goods and services are created or manufactured.
- **Costs of production:** The price of goods or services is determined by the cost of the resources used during production, according to this theory.
- **Labor economics:** This principle looks at workers and employers to understand patterns of wages, employment, and income.

The rules in microeconomics flow from a set of compatible laws and theorems rather than beginning with empirical study.

Microeconomics is the study of human action and interaction. The most common uses of microeconomics deal with individuals and firms that trade with one another, but its methods and insights can be applied to nearly every aspect of purposeful activity. Ultimately, microeconomics is about human choices and incentives.

Microeconomics argues that human actors are rational and that they use scarce resources to accomplish purposeful ends. The dynamic interaction between scarcity and choice helps economists discover what humans consider valuable. Exchange, demand, prices, profits, losses and competition arise when humans voluntarily associate with each other to achieve their separate ends.

1.1.7 UNDERSTANDING ECONOMIC THEORY

Economic theory is a set of principles and ideas that explain how economies function. It involves building models that use assumptions to describe economic behaviour. These models are selective because economic reality is complex and can't be fully represented. Economic theories may include:

- **Supply and demand:** A microeconomic theory that models how prices are determined.
- **Trade-offs:** A principle that states that to have more of one thing, an individual must have less of another.
- **Costs and benefits:** A theory based on rational choice, which states that people try to maximise the ratio of benefits to costs in their decisions.

Economic theory is different from other social and physical sciences because it claims to describe policies that will improve people's lives

Classical Economics

Those who favour classical economics assume that the economy is self-regulating and that any needs in an economy will be met by participants. In other words, there's no need for government intervention. People will allocate resources properly and efficiently. If there's a need in an economy, a company will start up to fill that need creating balance. Classical economists assume that people and companies will stimulate the economy, create growth, by spending and investment.

Neo-Classical Economics

Neo-classical economics is a broad theory that focuses on supply and demand as the driving forces behind the production, pricing, and consumption of goods and services. It emerged in around 1900 to compete with the earlier theories of classical economics.

Neo-classical economists assume that people make rational decisions when purchasing or investing in the economy. Prices are determined by supply and demand while there are no outside forces impacting prices. Consumers strive to maximise utility or their needs and wants. Maximising utility is a key tenet of rational choice theory, which focuses on how people achieve their objectives by making rational decisions. The theory holds that people, given the information they have, will opt for choices that provide the greatest benefit and minimise any losses.

Neoclassical economists believe the propensity for consumer need drives the economy and the business production that results to fill those needs. Any imbalances in an economy are believed to be corrected through competition, which restores equilibrium in the markets allocating resources properly.

NEO-CLASSICAL SUMMARY

- Classical economists assume that the most important factor in a product's price is its cost of production.
- Neoclassical economists argue that the consumer's perception of a product's value is the driving factor in its price.
- The difference between actual production costs and retail price is the economic surplus.
- Neoclassical economic theory can impact how businesses operate gov and financial institutions operate, as well as how governments regulate markets.
- Critics argue the theory doesn't account for other factors that impact consumer decisions, such as limited information, resource inequality, or emotional thinking

1.1.8 THE ASSUMPTIONS OF ECONOMISTS

The assumptions of economists are made to better understand consumer and business behaviour when making economic decisions. There are various economic theories to help explain how an economy functions and how to maximise growth, wealth, and employment.

However, the underlying themes of many theories centre around preferences, meaning what businesses and consumers prefer to have or prefer to avoid. Also, the assumptions usually involve the resources available or not available to fulfil the needs and preferences. The scarcity or abundance of resources is important in determining the choices that participants make in an economy.

Key Takeaways:

- The assumptions of economists are made to better understand consumer and business behaviour when making economic decisions.
- Some economists assume that people make rational decisions when purchasing or investing in the economy.
- Conversely, behavioural economists assume that people are emotional and can get distracted, thus influencing their decisions.
- Critics argue that assumptions in any economic model are often unrealistic and don't hold up in the real world.

1.1.9 POSITIVE vs NORMATIVE ECONOMICS

Positive economics and normative economics are two main subfields of economics that differ in their focus, methodology, and the type of statements they make

- Positive economics describes and explains various economic phenomena. Positive economic statements are based on facts and data and can be proven true or false. They use objective words like "is", "does", and "will". Positive economics is used to make short-term policy decisions.
- Normative economics focuses on value-based judgments about what the economy should be like, and how to improve economic development and wealth distribution. Normative economics is used in politics and policy making to build long-term policy goals.

To put it simply, positive economics is called the "what is" branch of economics. Normative economics, on the other hand, is considered the branch of economics that tries to determine people's desirability to different economic programs and conditions by asking what "should" be or what "ought" to be.

While positive economics is based on fact and cannot be approved or disapproved, normative economics is based on value judgments.

Most public policy is based on a combination of both positive and normative economics.

Positive Economics

Positive economics is a stream of economics that focuses on the description, quantification, and explanation of economic developments, expectations, and associated phenomena. It relies on objective data analysis, relevant facts, and associated figures. It attempts to establish any cause-and-effect relationships or behavioural associations which can help ascertain and test the development of economics theories.

Positive economics is objective and fact-based where the statements are precise, descriptive, and clearly measurable. These statements can be measured against tangible evidence or historical instances. There are no instances of approval-disapproval in positive economics.

Here's an example of a positive economic statement: **"Government-provided healthcare increases public expenditures."** This statement is fact-based and has no value judgment attached to it. Its validity can be proven (or disproven) by studying healthcare spending where governments provide healthcare.

Characteristics of positive economics

- **Objective:** Positive economics is based on facts, not opinions.
- **Predictive:** Positive economics uses generalisations to predict the consequences of changes in circumstances.
- **Measurable:** Positive economics statements are precise, descriptive, and can be measured against evidence or historical instances.
- **Reliable:** Positive economics can be used as a reliable source for decision-making by businesses and governments

Advantages	Disadvantages
Is easily verifiable because it is based on objective data	We can't always separate our emotions from the facts
Gives policymakers more power to make decisions	Economics isn't an exact science, so there are no fool-proof solutions or conclusions
Allows individuals to make wiser choices with their economic and financial lives	Policies and solutions that arise from positive economics don't affect everyone the same way
Is easily verifiable because it is based on objective data	We can't always separate our emotions from the facts

Normative Economics

Normative economics focuses on the ideological, opinion-oriented, prescriptive, value judgments, and "what should be" statements aimed toward economic development, investment projects, and scenarios. Its goal is to summarise people's desirability (or the lack thereof) to various economic developments, situations, and programs by asking or quoting what should happen or what ought to be.

Normative economics is subjective and value-based, originating from personal perspectives, feelings, or opinions involved in the decision-making process. Normative economics statements are rigid and prescriptive in nature. They often sound political or authoritarian, which is why this economic branch is also called "what should be" or "what ought to be" economics.

An example of a normative economic statement is: "The government should provide basic healthcare to all citizens." As you can deduct from this statement, it is value-based, rooted in personal perspective, and satisfies the requirement of what "should" be.



Normative economics is closely related to behavioural economics, the branch of psychology that examines the decisions of human actors. Some of the insights of behavioural economics include subtle ways to "nudge" consumers towards a desirable behaviour, without compelling or forbidding any one choice.

1.2. THEORY OF THE CONSUMER: DEMAND

In economics, the theory of consumer choice is related to demand as producer theory is related to supply.

	Consumer Choice Theory	Demand Theory
Focus	How consumers maximise the desirability of their consumption	The relationship between demand for goods and services and their prices
Explanation	Based on the idea that people's preferences can be inferred from their choices	Highlights the role of demand in price formation
Models	Used to represent demand patterns for an individual buyer	Forms the basis for the demand curve

- **Consumer choice theory** is a branch of microeconomics that analyses how consumers maximise their consumption while facing budget constraints.
- **Demand theory** is an economic principle that explains how the demand for goods and services relates to their prices.

1.2.1 IMPORTANT CONCEPTS IN MICRO ECONOMICS

Microeconomics is the study of what's likely to happen when individuals make choices in response to changes in incentives, prices, resources, or methods of production. These scenarios are also known as tendencies. Individuals are often grouped into microeconomic subgroups such as buyers, sellers, and business owners. **These groups create the supply and demand for resources, using money and interest rates as pricing mechanisms for coordination.**

Supply refers to the number of units of goods or services a supplier is willing and able to bring to the market for a specific price.

- The willingness and ability to avail products to the market are influenced by stock availability and the determiners driving the supply.
- A change in prices impacts the market equilibrium too.
- A price increase will result in more supplies, and a decrease will result in the opposite effect.

Demand relates to a consumer's desire to purchase goods and services and willingness to pay a specific price for them.

- An increase in the price of a good or service tends to decrease the quantity demanded.
- Likewise, a decrease in the price of a good or service will increase the quantity demanded.

These two economic forces influence each other; they are both important for the economy because they impact the prices of consumer goods and services within an economy.



Demand elasticity relates to how sensitive the demand for a product is as the price for it changes. For example, if there's a big change in demand due to a small change in price, demand elasticity is said to be high. Shoppers may choose attractive substitute products if the price for their usual product has increased somewhat. That could indicate high demand elasticity and is useful for businesses to know.

1.2.2 THE SUPPLY FUNCTION AND THE SUPPLY CURVE

The supply curve illustrates the correlation between the cost of a product or service and the quantity of it that is available. The supply curve is shown in a graph with the price on the left vertical axis and the quantity supplied on the horizontal axis.

The supply curve can be seen as a visual demonstration of how the law of supply and demand works. Prices increase when supply is low.

- A supply curve can often show if a commodity will experience a price increase or decrease based on demand, and vice versa.
- The supply curve is shallower (closer to horizontal) for products with more elastic supply and steeper (closer to vertical) for products with less elastic supply.
- The supply curve, along with the demand curve, are the key components of the law of supply and demand.

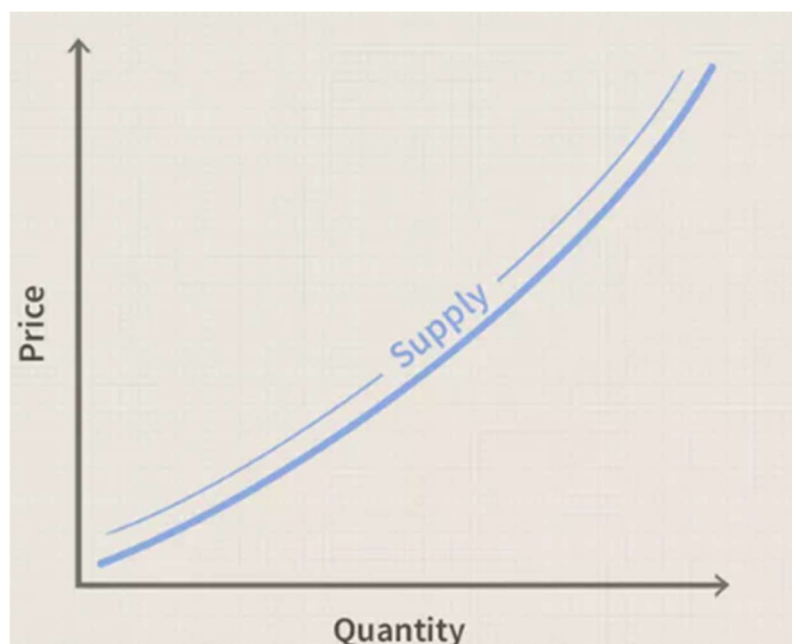
Prices influence producers' supply decisions. The supply function can be depicted as a positively sloped supply curve. That will move from left to right illustrating the law of supply.

- If all other factors are equal, a higher price will increase the producer's incentive to supply more goods.
- Higher prices increase the producer's profit, which is the excess of sales revenue over the cost of production.
- As the price of a good falls, its supply falls as well.

Therefore, there is a direct relationship between the price of a good and the amount of that good that will be supplied. The supply curve slopes upward. It tells the analyst the quantity that producers are willing to supply for each price when all other influences on producers' planned sales remain the same.

Note that this formulation implies that price is the independent variable, and quantity is the dependent variable. **In most disciplines, the independent variable appears on the horizontal or x-axis, but economics is an exception to this rule.**

The graph below displays the quantity associated with price in a supply table.

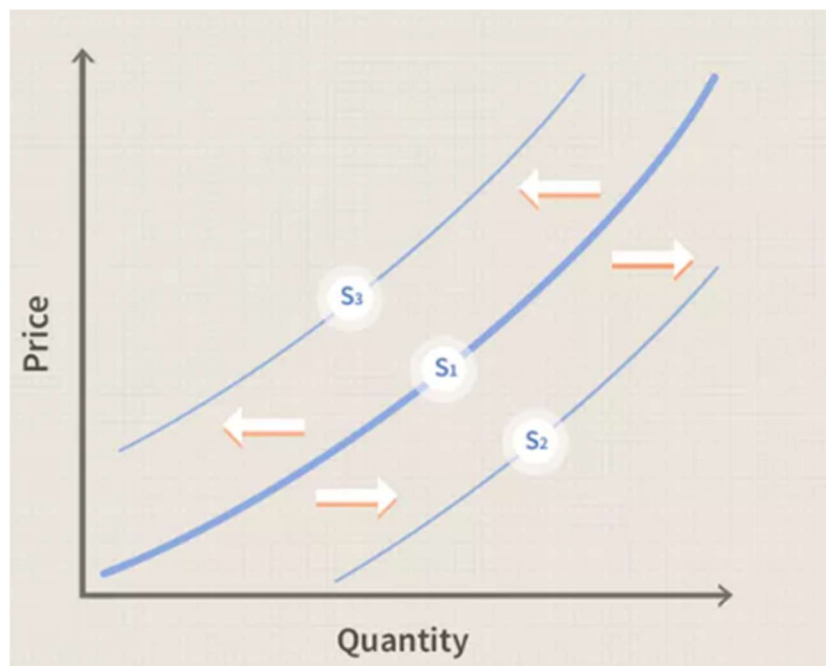


If a factor besides price or quantity changes, a new supply curve needs to be drawn. For example, say that more soybean farmers enter the market, clearing forests and increasing the amount of land devoted to soybean cultivation.

In this scenario, more soybeans will be produced even if the price remains the same, meaning that the supply curve itself shifts to the right (S_2) in the graph below. In other words, supply will increase

Other factors can shift the supply curve as well, such as a change in the price of production. If a drought causes water prices to spike, the curve will shift to the left (S_3). If the price of a substitute crop such as corn increases, farmers will shift to growing that instead, and the supply of soybeans will decrease (S_3).

If a new technology, such as a pest-resistant seed, increases yields, the supply curve will shift right (S_2). If the future price of soybeans is higher than the current price, the supply will temporarily shift to the left (S_3), since producers have an incentive to wait to sell.



WATCH the Video:

Demand and Supply Explained

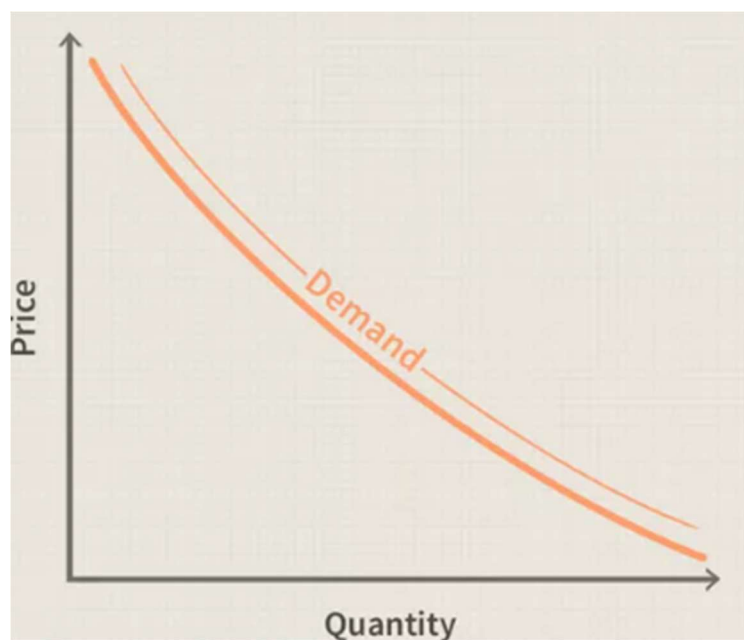
1.2.3 THE DEMAND FUNCTION AND THE DEMAND CURVE

The demand curve is a graphical representation of the relationship between the price of a good or service and the quantity demanded for a given period of time. In a typical representation, the price appears on the left vertical axis while the quantity demanded is on the horizontal axis.

- A demand curve is a graph that shows the relationship between the price of a good or service and the quantity demanded within a specified time frame.
- Demand curves can be used to understand the price-quantity relationship for consumers in a particular market, such as corn or soybeans.
- The demand curve generally slopes down from left to right, due to the law of demand while the quantity demanded drops as the price rises for the majority of goods.
- Changes in factors besides price and quantity can shift a demand curve to the right or left.

This curve generally moves downward from the left to the right. This movement expresses the law of demand, which states that as the price of a given commodity increases, the quantity demanded decreases as long as all else is equal.

Note that this formulation implies that price is the independent variable, and quantity is the dependent variable. In most disciplines, **the independent variable appears on the horizontal or x-axis**, but economics is an exception to this rule.



WATCH the Video:

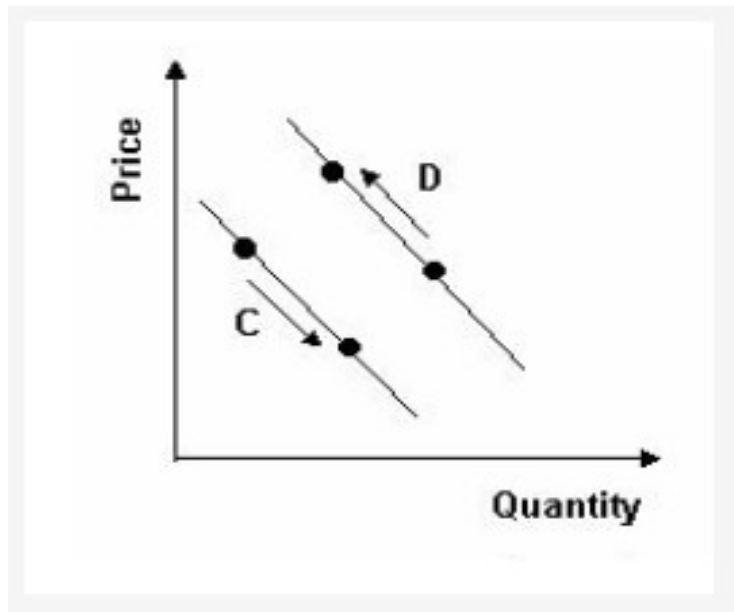
Introduction to Commodities

1.2.4 CHANGE IN DEMAND vs CHANGE IN QUANTITY DEMANDED

The demand curve isolates the impact of price on the amount of a product purchased.

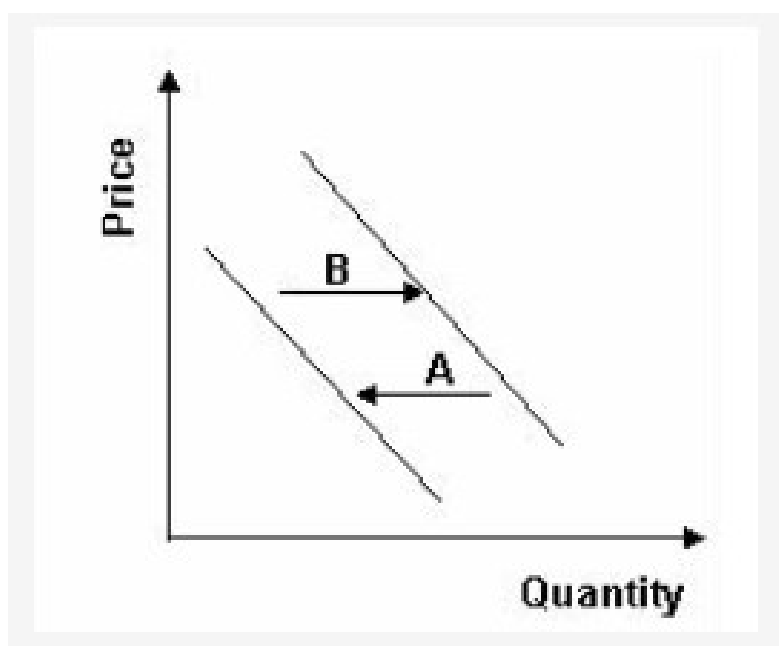
A **change in quantity demanded** (caused by price change ONLY) is a movement along a demand curve from one point to another.

- A change in price causes a movement along the demand curve. When price falls, the movement is downward and to the right.



Change In Demand

- Changes in other factors (anything other than price), such as income, tastes, expectations, and the prices of closely related goods, will shift the entire demand curve inwards or outwards. This is referred to as change in demand.



Elasticity of Demand and Supply

Price Elasticity of Demand (PED) is defined as the responsiveness of quantity demanded to a change in price. The demand for a product can be elastic or inelastic, depending on the rate of change in the demand with respect to the change in the price.

Demand is said to be elastic demand has a higher proportionate response to a smaller change in price. On the other hand, demand is inelastic when there is little movement in demand with a significant difference in price.

PED can be calculated by the percentage change in quantity divided by the percentage change in price.

Uses of price elasticity of demand include:

- Allows a firm or business to predict the change in total revenue with a projected change in price.
- Firms can charge different prices in different markets if elasticities differ in income groups. This practice is known as price discrimination. For example, airlines have segmented airplane seats into different classes – economy, business and first in order to charge the less price sensitive customer a higher price for premium seats.
- Allows a firm to decide how much tax to pass on to a consumer. If a product is inelastic, then the firm can force the customer to pay the tax. This is a common tactic used by cigarette manufacturers who pass on any health tax directly to the consumer.
- Enables the government to predict the impact of taxation policies on products.

Meaning of PED

- Elastic PES would mean that increases in the price will lead to disproportionately large increases in quantity supplied.
- Inelastic PES would mean that increases in the price will lead to disproportionately small increases in quantity supplied.
- Unit elastic PES would mean that increases in the price will lead to proportionately equal increases in quantity supplied

Types of Price Elasticity of Demand:

- Infinity: Perfectly elastic. Demand declines to zero
- Greater than 1. Elastic. Significant change in demand
- Equal to 1. Unitary elasticity. Equivalent percentage change in demand
- Less than 1. Inelastic. Insignificant change in demand
- Equal to 0. Perfectly inelastic. No change in demand

The price elasticity of supply (PES) is a measure of how responsive the quantity of a good or service supplied is to a change in its price

PES can be calculated by dividing the percentage change in quantity supplied by the percentage change in price.

The elasticity of supply, can be categorised as:

- **Elastic:** When the PES is greater than 1, indicating that the quantity supplied is highly responsive to price changes
- **Inelastic:** When the PES is less than 1, indicating that the quantity supplied is not very responsive to price changes
- **Perfectly elastic:** When the PES is infinite, meaning that the quantity supplied is unlimited at a given price.
- **Perfectly inelastic:** When the PES is 0, meaning that the quantity supplied does not change at all when the price changes.

Factors that can affect the elasticity of supply include:

- **Technology and innovation:** The use of new technologies and innovations can make supplies more elastic.
- **Flexibility:** The ability to substitute resources or quickly ramp up production in response to demand.
- **Time:** Supply is more elastic in the long run because there is more time to increase production.

1.2.5 MARKET EQUILIBRIUM

Market equilibrium occurs when quantities demanded equals the quantity supplied.

At the equilibrium point, the market price for a given good ensures that the quantity of goods supplied is equal to the number of goods demanded.

At this point, prices are perfectly set to interest consumers to purchase goods; at the same time, ensuring that companies produce neither too much nor too little product.

Both economists and companies analyse the relationship between supply and demand when making strategic product decisions. The assumption behind a market economy is that supply and demand are the best determinants for an economy's growth and health.

1.2.6 GOVERNMENT POLICY AND MICROECONOMICS

Government policy impacts microeconomics by influencing the incentives and inputs that shape individual decisions. Microeconomics is the study of how individuals make choices in response to changes in prices, incentives, resources, or production. The government can impact microeconomics through policies and regulations that change taxes, fiscal policy, interest rates, tariffs, subsidies, and more

- Government policy influences microeconomics by implementing incentives and inputs that shape individual decisions.
- Changes to tax policy, fiscal policy, interest rate policy, regulations, tariffs, and subsidies affect individual choices.
- Microeconomics studies the interaction of supply and demand when individuals make choices in response to changes in incentives, prices, resources, or production

Intentional and Unintentional Consequences:

- Sometimes the impacts of government policy are intentional. The government might provide a subsidy to farmers to make their businesses more profitable and encourage farm production.
- Conversely, the government might put a tax on cigarettes and alcohol to discourage behaviour that it doesn't approve of. Other impacts are unintentional.

1.2.7 PRICE CEILING AND PRICE FLOOR

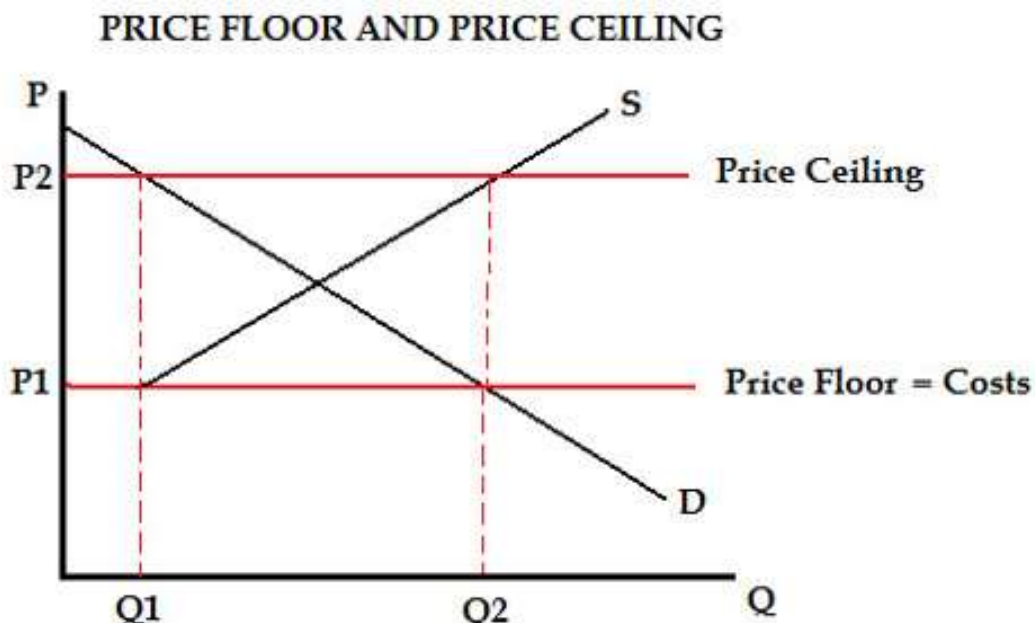
One keyway for a government to intervene in the market is through price controls. A price control is a regulation that prevents a price from rising and/or falling beyond a certain level.

Price ceiling

Price Ceilings are maximum prices set by the government for particular goods and services that they believe are being sold at too high of a price and thus consumers need some help purchasing them. Price ceilings only become a problem when they are set below the market equilibrium price. When the ceiling is set below the market price, there will be excess demand or a supply shortage.

Price floor

Price Floors are minimum prices set by the government for certain commodities and services that it believes are being sold in an unfair market with too low of a price and thus their producers deserve some assistance. Price floors are only an issue when they are set above the equilibrium price, since they have no effect if they are set below market clearing price. When they are set above the market price, then there is a possibility that there will be an excess supply or a surplus



Impact of a Price Ceiling (Maximum Price)

The big advantage of a price ceiling is the limit on costs for the consumer. It keeps things affordable and prevents price-gouging and producers/suppliers from taking unfair advantage of them.

Ceilings can mitigate the pain of higher prices until supply returns to normal levels if it's just a temporary shortage that's causing rampant inflation. Price ceilings can also stimulate demand and encourage spending.

Price ceilings have their advantages in the short term, but they can become a problem if they continue for too long or when they're set too far below the market equilibrium price when the quantity demanded equals the quantity supplied. Demand can skyrocket when this happens, leading to shortages in supply.

Something will have to give if the prices that producers are allowed to charge are too out of line with their production costs and business expenses. They may have to cut corners, reduce quality, or charge higher prices on other products.

They may have to discontinue offerings or not produce as much, causing more shortages. Some may even be driven out of business if they can't realise a reasonable profit on their goods and services.

Types of Price Ceilings

Governments can implement several types of price ceilings depending on the good that's being regulated and the entity that's doing the regulating.

- **Absolute price ceiling:** This is a fixed limit on the price that can be charged for a good or service. The price can't go higher than this limit. A government can vote or decide to periodically change the absolute price ceiling, but these ceiling limit changes are often infrequent enough to uphold the usefulness of the regulation.
- **Relative price ceiling:** This is a limit on the price of a good or service in relation to another good or service. A government might set a relative price ceiling on rent in relation to the average income of renters in a particular area. The government would closely monitor the independent variable in this case to better understand how the dependent variable's price ceiling fluctuates.
- **Per unit price ceiling:** This is a limit on the price of a good or service per unit. A government might set a per-unit price ceiling on the price of gasoline. This is similar to the absolute price ceiling although a consumer could theoretically spend as much as they wanted if they were to buy higher quantities.
- **Periodic price ceiling:** This is a limit on the price of a good or service for a specific period. A government might set a periodic price ceiling on electricity during the summer months when demand is high. Governments may enact periodic price ceilings during times of emergency.
- **Selective price ceiling:** This is a limit on the price of a specific type of good or service. A government might set a selective price ceiling on prescription drugs to make them more affordable for consumers.

Summary: Price Ceilings

- A price ceiling is a type of price control that's usually government-mandated and sets the maximum amount a seller can charge for a good or service.
- Price ceilings are typically imposed on consumer staples like food, gas, or medicine, often after a crisis or particular event sends costs skyrocketing.
- The opposite of a price ceiling is a price floor, a point below which prices can't be set.
- Price ceilings make staples affordable for consumers in the short term, but they often carry long-term disadvantages such as shortages, extra charges, or lower-quality products.
- Economists worry that price ceilings cause a deadweight loss to an economy, making it more inefficient.

Impact of a Price Floor (Minimum Price)

Price floors are sometimes called "price supports", because they support a price by preventing it from falling below a certain level.

We will use an example of the effects of a government policy that imposes a price floor (a minimum price) above the equilibrium price for a product such as eggs.

Price floors can have several consequences, including:

- **Surplus:** When a price floor is set above the market equilibrium price, there is an excess supply of the product. This is because the price floor prevents prices from falling to the equilibrium level, where quantity demanded equals quantity supplied.
- **Reduced demand:** Consumers may reduce their purchases, switch to substitutes, or drop out of the market entirely.
- **Producer surplus:** The effect of mandating a higher price transfer some of the consumer surplus to producer surplus.
- **Deadweight loss:** The price moving upward from the equilibrium price creates a deadweight loss.
- **Market failure:** A price floor may lead to market failure if the market is not able to allocate scarce resources in an efficient manner.
- **Unemployment:** A surplus can lead to unemployment, which the government addresses through various compensation and support programs as well as work programs.
- **Perishable goods:** A surplus of perishable goods can be disastrous to the manufacturer if their products spoil.
- **Backdoor arrangements:** Rent controls can create arrangements like requirements that tenants rent items they don't want or outright bribes.

Governments often set price floors in agricultural markets to assist farmers.

Price floors also create winners and losers. Those producers that can sell at the higher price benefit, while consumers lose since they must pay a higher price. At this higher price, there are fewer consumers of the good or service

1.2.8 FACTORS THAT INFLUENCE COMPETITION IN MICROECONOMICS

Competition in economics is when different firms compete to obtain limited goods by varying their marketing mix. Competition can influence a company's marketing plan, including pricing strategy, product development, and promotional activities

Factors that influence competition in microeconomics include:

- **Product features:** The quality and features of a product affect competition.
- **Number of sellers:** The number of sellers in a market affects competition.
- **Barriers to entry:** Barriers to entry can prevent new competitors from entering a market, which can limit competition.
- **Information availability:** The availability of information affects competition.
- **Location:** The location of a business affects competition.
- **Government policy:** Government policy can affect competition.
- **Market size structure:** The structure of the market can affect competition.
- **Customer availability and financial capability:** The availability and financial capability of customers can affect competition.

Each factor hinges on the availability or attractiveness of substitutes and, when no alternatives exist and the company is a single seller of a unique product, a monopoly exists and there is zero competition. In the next section will take a look at the different market structures.

1.3. MARKET STRUCTURES

In economics, market structure is a way to categorise markets by the number of firms producing similar products, and other characteristics

1.3.1. THE FIRM AND MARKET STRUCTURES

As different market structures result in different sets of choices facing a firm's decision makers, an understanding of market structure is a powerful tool in analysing issues such as a firm's pricing of its products and, more broadly, its potential to increase profitability. In the long run, a firm's profitability will be determined by the forces associated with the market structure within which it operates.

In a highly competitive market, long- run profits will be driven down by the forces of competition. In less competitive markets, large profits are possible even in the long run; in the short run, any outcome is possible.

Therefore, understanding the forces behind the market structure will aid the financial analyst in determining firms' short- and long- term prospects.

The four-market structure are:

- **Perfect competition:** A market with many small companies selling the same product at similar prices. There are few barriers to entry, and companies have little influence over the market.
- **Monopolistic Competition:** Refers to an imperfectly competitive market with the traits of both the monopoly and competitive market.
- **Oligopoly:** A market with a few firms and barriers to entry, where there is often high competition between firms. Decisions made by one oligopoly can directly affect another.
- **Monopoly:** A market with only one seller, who has complete control over the market and can set any price. Consumers have no other option and must pay the price set by the seller.

At one end of the spectrum is perfect competition, in which many firms produce identical products, and competition forces them all to sell at the market price. At the other extreme, we have monopoly, where only one firm is producing the product. In between are monopolistic competition (many sellers and differentiated products) and oligopoly (few firms that compete in a variety of ways). Each market structure has its own characteristics and implications for firm strategy, and we will examine each in turn.

1.3.2. THE STRUCTURES

Perfect Competition

Perfect competition occurs when there is a large number of small companies competing against each other. They sell similar products (homogeneous), lack price influence over the commodities, and are free to enter or exit the market.

Consumers in this type of market have full knowledge of the goods being sold. They are aware of the prices charged on them and the product branding. In the real world, the pure form of this type of market structure rarely exists. However, it is useful when comparing companies with similar features. This market is unrealistic as it faces some significant criticisms described below.

- **No incentive for innovation:** In the real world, if competition exists and a company holds a dominant market share, there is a tendency to increase innovation to beat the competitors and maintain the status quo. However, in a perfectly competitive market, the profit margin is fixed, and sellers cannot increase prices, or they will lose their customers.
- **There are very few barriers to entry:** Any company can enter the market and start selling the product. Therefore, incumbents must stay proactive to maintain market share.

Characteristics perfect competition:

- **A Large and Homogeneous Market:** A large population of both buyers and sellers ensures that supply and demand remain constant in this market. Buyers can easily substitute products made by one firm for others.
- **Perfect Information Availability:** The availability of free and perfect information in a perfectly competitive market ensures that each firm can produce its goods or services at the same rate and with the same production techniques as another in the market.

- **Absence of Controls:** The entry and exit of firms are unregulated and this frees them up to spend on labour and capital assets without restrictions and adjust their output in relation to market demands.
- **Cheap and Efficient Transportation:** Companies don't incur significant costs to transport goods in this type of market. This helps reduce the product's price and cuts back on delays in transporting goods.

The advantage and disadvantage of perfect competition are indicated in the table below:

Advantage	Disadvantage
▪ Provides a convenient framework for modelling market activity. ▪ Demonstrates how producers are incentivised to provide lower prices ▪ Provides a convenient framework for modelling market activity.	▪ The perfect competition model does not always reflect real-world market conditions. ▪ The model does not account for geographical differences or variations between products. ▪ The perfect competition model does not always reflect real-world market conditions. ▪ The model does not account for how producers benefit from economies of scale

Monopolistic Competition

It differs from perfect competition in that products are not identical. Each firm differentiates its product(s) from those of other firms through some combination of differences in product quality, product features, and marketing.

When comparing monopolistic competition in the short term and long term, there are two distinct aspects that are observed. In the short term, the monopolistic company maximises its profits and enjoys all the benefits as a monopoly.

The company initially produces many products as the demand is high. Therefore, its Marginal Revenue (MR) corresponds to its Marginal Cost (MC). However, MR diminishes over time as new companies enter the market with differentiated products affecting demand, leading to less profit.

Oligopoly

Monopolistic competition is a type of market structure where many companies are present in an industry, and they produce similar but differentiated products. None of the companies enjoy a monopoly, and each company operates independently without regard to the actions of other companies. The market structure is a form of imperfect competition.

Barriers to entry are high, often because economies of scale in production or marketing led to very large firms. Demand can be more or less elastic than for firms in monopolistic competition. The automobile market is dominated by a few very large firms and can be characterised as an oligopoly. The product and pricing decisions of Toyota certainly affect those of Ford and vice versa. Automobile makers compete based on price, but also through marketing, product features, and quality, which is often signalled strongly through brand name. The oil industry also has a few dominant firms, but their products are very good substitutes for each other.

Monopoly

A monopoly market is characterised by a single seller of a product with no close substitutes. This fact alone means that the firm faces a downward-sloping demand curve (the market demand curve) and has the power to choose the price at which it sells its product. High barriers to entry protect a monopoly producer from competition. One source of monopoly power is the protection offered by copyrights and patents.

Another possible source of monopoly power is control over a resource specifically needed to produce the product. Most frequently, monopoly power is supported by government. A natural monopoly refers to a situation where the average cost of production is falling over the relevant range of consumer demand. In this case, having two (or more) producers would result in a significantly higher cost of production and be detrimental to consumers. Examples of natural monopolies include the electric power and distribution business and other public utilities. When privately owned companies are granted such monopoly power, the price they charge is often regulated by government as well.

In a situation where companies mutually compete, they may create agreements to share the market by restricting production, leading to supernormal profits. This holds if either party honours the Nash equilibrium state, and neither is tempted to engage in the prisoner's dilemma. In such an agreement, they work like monopolies. The collusion is referred to as cartels.

1.3.3. INVESTMENT CHARACTERISTICS

The table in Characteristics of Market Structures shows the key features of each market structure.

We can analyse where an industry falls along this spectrum by examining the following five factors:

Number of firms and their relative sizes.

1. Degree to which firms differentiate their products.
2. Bargaining power of firms with respect to pricing.
3. Barriers to entry into or exit from the industry.
4. Degree to which firms compete on factors other than price.

	Perfect Competition	Monopolistic Competition	Oligopoly	Monopoly
Number of sellers	Many firms	Many firms	Few firms	Single firm
Barriers to entry	Very low	Low	High	Very high
Nature of substitute products	Very good substitutes	Good substitutes but differentiated	Very good substitutes or differentiated	No good substitutes
Nature of competition	Price only	Price, marketing, features	Price, marketing, features	Advertising
Pricing power	None	Some	Some to significant	Significant

1.4. BEHAVIOURAL ECONOMICS

Behavioural Economics is the study of psychology as it relates to the economic decision-making processes of individuals and institutions. Behavioural economics is often related with normative economics. It draws on psychology and economics to explore why people sometimes make irrational decisions, and why and how behaviour diverges from the predictions of economic models.

The two most important questions in this field are:

- Are economists' assumptions of utility or profit maximisation good approximations of real people's behaviour?
- Do individuals maximise subjective expected utility?

Behavioural economics is often related with normative economics. Normative economics is a perspective on economics that reflects normative, or ideologically prescriptive judgments toward economic development, investment projects, statements, and scenarios.

Behavioural economics draws on psychology and economics to explore why people sometimes make irrational decisions, and why and how their behaviour does not follow the predictions of economic models. Decisions such as how much to pay for a cup of coffee, whether to go to graduate school, whether to pursue a healthy lifestyle, how much to contribute towards retirement, etc. are the sorts of decisions that most people make at some point in their lives. Behavioural economics seeks to explain why an individual decided to go for choice A, instead of choice B.

Summary: Behavioural Economics

- Behavioural economics is the study of psychology that analyses the economic decisions people make.
- Factors that affect behaviour include bounded rationality, choice architecture, cognitive biases, discrimination, and herd mentality.
- Behaviour economics is crafted around many principles including framing, heuristics, loss aversion, and the sunk-cost fallacy.
- Companies use information from behavioural economics to price their goods, craft their commercials, and package their products.

1.4.1. UNDERSTANDING BEHAVIOURAL ECONOMICS

In an ideal world, people would always make optimal decisions that provide them with the greatest benefit and satisfaction.

In economics, rational choice theory states that when humans are presented with various options under the conditions of scarcity, they would choose the option that maximises their individual satisfaction. This theory assumes that people, given their preferences and constraints, can make rational decisions by effectively weighing the costs and benefits of each option available to them. The final decision made will be the best choice for the individual.

The rational person has self-control and is unmoved by emotions and external factors and, hence, knows what is best for himself. Alas behavioural economics explains that humans are not rational and are incapable of making good decisions.

1.4.2. APPLICATION IN CAPITAL MARKETS

Behavioural economics can be applied to is behavioural finance, which seeks to explain why investors make rash decisions when trading in the capital markets.

Behavioural finance, a sub-field of behavioural economics, proposes that psychological influences and biases affect the financial behaviours of investors and financial practitioners. Moreover, influences and biases can be the source for explanation of all types of market anomalies and specifically market anomalies in the stock market, such as severe rises or falls in stock price.

Behavioural finance provides insight into how emotional biases and cognitive errors may influence individuals' perceptions and investment decisions. Consequently, knowledge of behavioural biases may help in understanding client goals, in constructing investment portfolios, and in identifying inconsistencies in investment decision making. Behavioural finance also provides insights into issues such as market anomalies.

The idea that psychology drives stock market movements flies in the face of established theories that advocate the notion that financial markets are efficient. Proponents of the efficient market hypothesis (EMH), for instance, claim that any new information relevant to a company's value is quickly priced by the market. As a result, future price moves are random because all available (public and some non-public) information is already discounted in current values.

However, for anyone who has been through the Internet bubble and the subsequent crash, the efficient market theory is hard to swallow. Behaviourists explain that, rather than being anomalies, irrational behaviour is commonplace. In fact, researchers have regularly reproduced examples of irrational behaviour outside of finance using very simple experiments.

1.4.3. BEHAVIOURAL FINANCE

Behavioural finance, a subfield of behavioural economics, proposes that psychological influences and biases affect the financial behaviours of investors and financial practitioners. Moreover, influences and biases can be the source for the explanation of all types of market anomalies and specifically market anomalies in the stock market, such as severe rises or falls in stock price.

Within behavioural finance, it is assumed that financial participants are not perfectly rational and self-controlled but rather psychologically influential with somewhat normal and self-controlling tendencies. Financial decision-making often relies on the investor's mental and physical health. As an investor's overall health improves or worsens, their mental state often changes. This impacts their decision-making and rationality towards all real-world problems, including those specific to finance.

One of the key aspects of behavioural finance studies is the influence of biases. Biases can occur for a variety of reasons. Biases can usually be classified into one of five key concepts. Understanding and classifying different types of behavioural finance biases can be very important when narrowing in on the study or analysis of industry or sector outcomes and results.



Familiarity bias can occur in so many ways. You may resist investing in a specific company because of what industry it is in, where it operates, what products it sells, who oversees the management of the company, who its clientele base is, how it performs its marketing, and how complex its accounting is.

Behavioural finance typically encompasses five main concepts:

- **Mental accounting:** Mental accounting refers to the propensity for people to allocate money for specific purposes.
- **Herd behaviour:** Herd behaviour states that people tend to mimic the financial behaviours of much of the herd. Herding is notorious in the stock market as the cause behind dramatic rallies and sell-offs.
- **Emotional gap:** The emotional gap refers to decision-making based on extreme emotions or emotional strains such as anxiety, anger, fear, or excitement. Oftentimes, emotions are a key reason why people do not make rational choices.
- **Anchoring:** Anchoring refers to attaching a spending level to a certain reference. Examples may include spending consistently based on a budget level or rationalising spending based on different satisfaction utilities.
- **Self-attribution:** Self-attribution refers to a tendency to make choices based on overconfidence in one's own knowledge or skill. Self-attribution usually stems from an intrinsic knack in a particular area. Within this category, individuals tend to rank their knowledge higher than others, even when it objectively falls short.



Behavioural finance is exploited through credit card rewards, as consumers are more likely to be willing to spend points, rewards, or miles as opposed to paying for transactions with direct cash.

Traditional Financial Theory

The traditional financial theory believes that everyone will take all available information when making investment decisions.

In order to better understand behavioural finance, let's first look at traditional financial theory. Traditional finance includes the following beliefs:

- Both the market and investors are perfectly rational
- Investors truly care about utilitarian characteristics
- Investors have perfect self-control
- They are not confused by cognitive errors or information processing errors

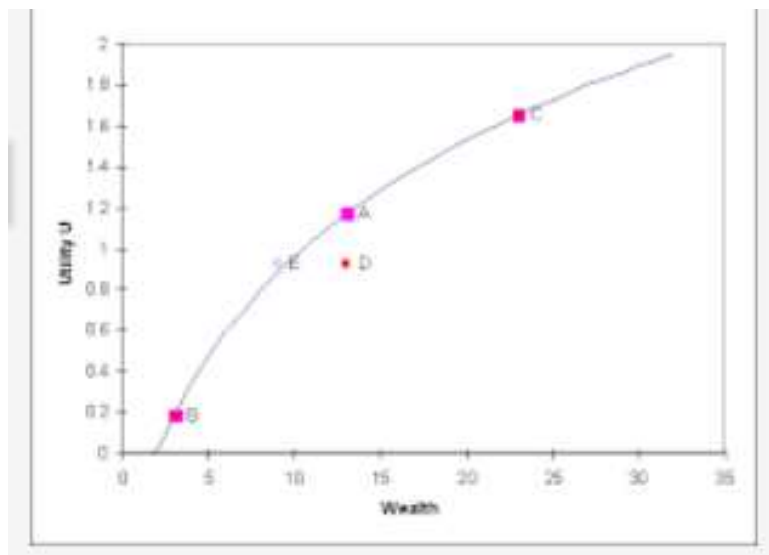
Behavioural Finance Theory

Now let's compare traditional financial theory with behavioural finance. Traits of behavioural finance are:

- Investors are treated as “normal” not “rational”
- They actually have limits to their self-control
- Investors are influenced by their own biases
- Investors make cognitive errors that can lead to wrong decisions

1.4.4. TRADITIONAL FINANCE: INVESTOR DECISION-MAKING

- It is a normative approach, which means that it explains how individuals acting rationally will invest to maximise utility.
- It's based on the concept of Utility, which is "the level of relative satisfaction received from the consumption of a good or service."
- An individual is a Rational Economic Man (REM) who is:
 - Perfectly rational
 - Perfectly self-interested
 - Has access to perfect information
- Traditional finance assumes that investors are risk averse, which means that they derive less and less utility from each additional unit of wealth as they grow wealthier.



1.4.5. BEHAVIOURAL FINANCE: INVESTOR DECISION MAKING

Behavioural finance examines investor behaviour to understand how people make decisions, individually and collectively. Behavioural finance does not assume that investors always act rationally, but instead that people can be negatively affected by behavioural biases.

Market efficiency does not require all market participants to act rationally as long as the market acts rationally in aggregate. If the market is able to quickly adjust for irrationality, then behavioural finance does not necessarily contradict market efficiency. However, if the market allows its participants to earn abnormal returns from the irrationality of others, then the market cannot be efficient.

- **Loss Aversion Bias:** People have a tendency to dislike losses more than they like comparable gains. This may help to explain under-reaction and over-reaction market anomalies.
- **Herding Bias:** Market participants tend to trade along with other investors, while potentially ignoring their own private information or analysis. This bias may also serve as a possible explanation for the under-reaction and overreaction market anomalies.
- **Information Cascades:** Similar to herding, information cascade is the transmission of information from those participants who act first and whose decisions influence the decisions of others. As investors base the decisions on the actions of other investors acting before them, stock returns may be serially correlated and lead to over-reaction anomalies. Research has shown information cascades to be greater for companies with poor quality information.
- **Overconfidence Bias:** People tend to overestimate their ability to accurately determine intrinsic values, and may not process information appropriately as a result, which ultimately leads to mispriced securities. This mispricing has been shown to mainly take place in higher-growth companies, whose prices react slowly to new information.

Other Biases

Other biases include representativeness (overweighting current situation in making decisions), mental accounting (separately accounting for different investments and individual security gains/losses), conservatism (maintain prior views or forecasts despite new information), and narrow framing (viewing issues in isolation and responding based on how issues are posed).

1.4.6. UTILITY THEORY: THE THEORY OF CONSUMER BEHAVIOUR

Utility theory is an economic theory that explains consumer behaviour by assuming that people can consistently rank their choices based on their preferences

Utility Theory is associated with the Tradition Finance perspective.

- Traditional finance assumes that individuals are Rational Economic Men, who are perfectly rational, perfectly self-interested, and have access to perfect information.
- Stemming from these assumptions is the Utility Theory,
 - **Utility:** The satisfaction or pleasure a person gets from consuming a good or service
 - **Total utility:** The overall satisfaction from consuming a specific quantity of a good or service
 - **Marginal utility:** The additional satisfaction from consuming one more unit of a good or service
- Individuals make decisions based on their ongoing process of calculating probabilities of expected utility for option A vs. option B (yes, that's what we are apparently doing every time we make a decision).

1.4.7. THE EFFICIENT MARKET HYPOTHESIS

The efficient market hypothesis (EMH), alternatively known as the efficient market theory, is a hypothesis that states that share prices reflect all available information and consistent alpha generation is impossible.

According to the EMH, stocks always trade at their fair value on exchanges, making it impossible for investors to purchase undervalued stocks or sell stocks for inflated prices.¹ Therefore, it should be impossible to outperform the overall market through expert stock selection or market timing. The only way an investor can obtain higher returns is by purchasing riskier investments.

- The efficient market hypothesis (EMH) or theory states that share prices reflect all the information available.
- According to the EMH, stocks trade at their fair market value on exchanges.
- Proponents of EMH posit that investors benefit from investing in a low-cost, passive portfolio.
- Opponents of EMH believe that it is possible to beat the market and that stocks can deviate from their fair market values.
- The EMH carries essentially the same implication for investors as the Random Walk Theory – is based on a number of assumptions about securities markets and how they function.

Three forms of efficient-market hypothesis:

The efficient-market hypothesis says that financial markets are effective in processing and reflecting all available information with little or no waste, making it impossible for investors to consistently outperform the market based on information already known to the public. One area of debate is how strong the efficient-market hypothesis is. In 1970, Fama wrote another paper that explored the idea of market efficiency in more depth, noting that it seemed to take three forms:

- **Weak-form efficiency:** In this form, market prices reflect all past trading information, such as historical prices and trading volumes. According to weak-form efficiency, technical analysis (the study of past price and volume data) cannot consistently generate excess returns because this information is already reflected in stock prices.
- **Semi-strong-form efficiency:** This idea says that all publicly available information, including news and past trading data, is fully reflected in stock prices. As a result, neither technical analysis nor fundamental analysis (the study of financial statements and economic factors) can consistently beat the market, because all available information is already incorporated into prices.
- **Strong-form efficiency:** The most robust version of the efficient-market hypothesis contends that all information, public and private, is fully reflected in stock prices. In other words, no individual or group of investors possesses information that can consistently yield superior returns. This form of efficiency suggests that insider trading is futile in the long run, as insider information is also reflected in stock prices.

Criticisms and Limitations

despite its significance, the efficient-market hypothesis is not without criticisms and limitations. Some critics argue that several factors prevent markets from being perfectly efficient, including:

- **Behavioural biases** - errors in judgment, decision-making, and thinking when evaluating information.
- **Information asymmetry** - where one person has more or better information than someone else.
- **Market frictions** - anything that interferes with market transactions, including transaction costs, taxes, regulation, and information glitches.

Critics of the EMH point to market bubbles, crashes, and persistent anomalies as evidence against strong market efficiency. An entire field of finance, behavioural economics, has developed to explore how market participants are inefficient.

Another criticism of the efficient-market hypothesis is that certain valuation anomalies persist, even though the hypothesis says they shouldn't. One is that small companies tend to outperform larger ones; another is that value stocks tend to outperform those with higher price-to-earnings (P/E) ratios. In 1992, Fama and French published a paper showing that those anomalies were real and should be incorporated into financial valuation models.

The efficient-market hypothesis remains a cornerstone of financial theory and has had a profound influence on investment strategies, portfolio management, and the understanding of financial markets. Although its three forms provide an accepted framework for thinking about market efficiency, the debate about its validity continues.



READ the Article:

The Efficient Market Hypothesis

1.5. MACRO ECONOMICS

Macro-economics is the study of the whole economy by considering the effects of the following factors on economic activities:

- Inflation
- Economic growth
- Unemployment
- Interest rates
- Exchange rates

In addition, macroeconomists develop models explaining the relationships between these factors.

These models, and the forecasts they produce, are used by government entities to aid in constructing and evaluating economic, monetary, and fiscal policy. Businesses use the models to set strategies in domestic and global markets, and investors use them to predict and plan for movements in various asset classes.

Macro-economic variables have the following effects on investments:

- Some firms are more sensitive to economic growth and inflation rates
- Bond prices change inversely with interest rates
- Multinational firms are affected by foreign exchange rates
- Import and export firms are affected by changes in foreign exchange rates

1.5.1. LIMITS OF MACROECONOMICS

It is also important to understand the limitations of economic theory. Theories are often created in a vacuum and lack specific real-world details like taxation, regulation, and transaction costs. The real world is also decidedly complicated and includes matters of social preference and conscience that do not lend themselves to mathematical analysis



It is common to find the phrase *ceteris paribus*, loosely translated as "all else being equal" in economic theories and discussions. Economists use this phrase to focus on specific relationships between variables being discussed, while assuming all other variables remain fixed.

Even with the limits of economic theory, it is important and worthwhile to follow significant macroeconomic indicators like GDP, inflation, and unemployment. This is because the performance of companies, and by extension their stocks, is significantly influenced by the economic conditions in which the companies operate

1.5.2. ECONOMIC GROWTH

Economic growth refers to an increase in the size of a country's economy over a period of time. The size of an economy is typically measured by the total production of goods and services in the economy, which is called gross domestic product (GDP).

Economic growth can be measured in 'nominal' or 'real' terms. Nominal economic growth refers to the increase in the dollar value of production over time. This includes changes in both the volume of production and the prices of goods and services produced. Economists normally talk about real economic growth – that is, increases in the volume produced only, which takes away the effect of prices changing. This is because it better reflects how much a country is producing at a given time, compared with other points in time.

Preconditions for Growth

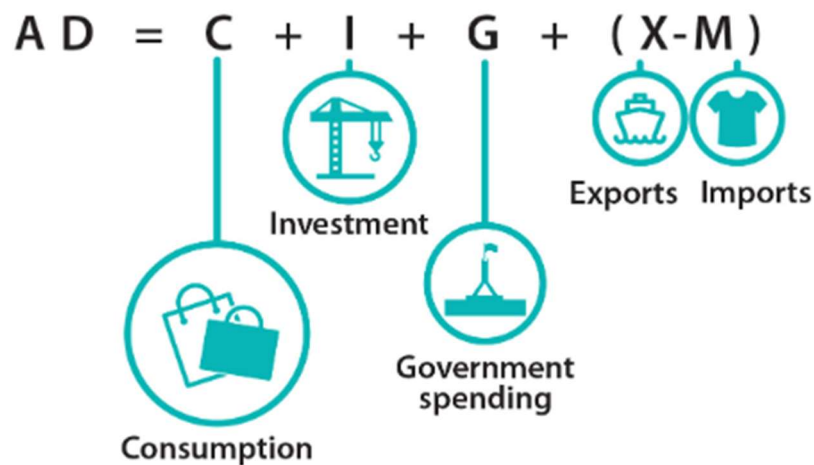
Savings and investment	Positively correlated with economic development. For countries to grow, private and public sector investment must provide a sufficient level of capital per worker. If a country has insufficient domestic savings, it must attract foreign investment in order to grow.
Financial markets and intermediaries	Augment economic growth by efficiently allocating resources in several ways. First, financial markets determine which potential users of capital offer the best returns on a risk-adjusted basis. Second, financial instruments are created by intermediaries that provide investors with liquidity and opportunities for risk reduction. Finally, by pooling small amounts of savings from investors, intermediaries can finance projects on larger scales than would otherwise be possible Some caution is in order, however. Financial sector intermediation may lead to declining credit standards and/or increases in leverage, increasing risk but not economic growth.
Political stability	Rule of law, and property rights environment of a country also influence economic growth. Countries that have not developed a system of property rights for both physical and intellectual property will have difficulty attracting capital. Similarly, economic uncertainty caused by wars, corruption, and other disruptions poses unacceptable risk to many investors, reducing potential economic growth.
Investment in human capital	The investment in skills and well-being of workers, is thought to be complementary to growth in physical capital. Consequently, countries that invest in education and health care systems tend to have higher growth rates. Developed countries benefit the most from post-secondary education spending, which has been shown to foster innovation. Less-developed countries benefit the most from spending on primary and secondary education, which enables the workforce to apply the technology developed elsewhere.
Tax and regulatory systems	Need to be favourable for economies to develop. All else equal, the lower the tax and regulatory burdens, the higher the rate of economic growth. Lower levels of regulation foster entrepreneurial activity (startups), which have been shown to be positively related to the overall level of productivity.
Free trade and unrestricted capital flows	Are also positively related to economic growth. Free trade promotes growth by providing competition for domestic firms, thus increasing overall efficiency and reducing costs. Additionally, free trade opens up new markets for domestic producers. Unrestricted capital flows mitigate the problem of insufficient domestic savings as foreign capital can increase a country's capital, allowing for greater growth. Foreign capital can be invested directly in assets such as property, physical plant, and equipment (foreign direct investment), or invested indirectly in financial assets such as stocks and bonds.

1.5.3. COMPONENTS OF GDP

GDP can be measured from an expenditure or income perspective but the two should be equal. The components of GDP from an expenditure perspective are:

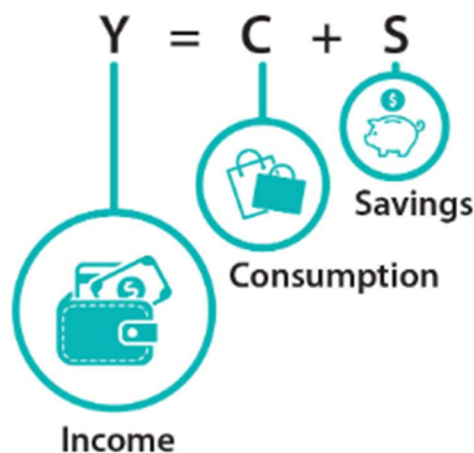
- Consumption spending by individuals (C)
- Investments in assets by businesses (I)
- Government spending on goods and services (G)
- Net exports (exports minus imports) (X-Z)

Aggregate demand (AD), like GDP (total value of expenditure by consumers, businesses and governments on final goods and services.), refers to the total level of spending in the economy. Consequently, when aggregate demand is measured it is the same as GDP (total value of expenditure by consumers, businesses and governments on final goods and services). Aggregate demand (AD) includes household spending (also called consumption, C), investment by businesses and households (I), spending by the government (G) and net spending from overseas (X-M).



Household consumption (C) refers to spending by households on things like rent, groceries and utilities. It makes up the largest share of aggregate demand. The level of consumption by each household is largely dependent on their level of income (Y). Household income that is not spent, is saved (S).

When household income increases, household spending usually increases as well. The amount of extra consumption for an extra rand of income is called the **Marginal Propensity to Consume (MPC)**.



1.5.4. AGGREGATE OUTPUT AND INCOME

Gross domestic product: The most common measure of aggregate output is the market value of all final goods and services produced in a country during the course of the year.

Aggregate income: is the total income of the factors of production (land, labour, capital) from producing goods and services in the economy during the year.

Real vs Nominal GDP:

- **Nominal GDP** – When the total value of goods and services are calculated using current prices. Nominal indicates values measured at current prices.
- **Real GDP** – Expresses values of economic production in terms of prices for an arbitrary base year. Real GDP measures the quantities of goods and services and does not change because the prices have changed.

1.5.5. Growth Rate

The media often talk about the economy's growth rate and the growth rate of real GDP.

A growth rate is defined as the percentage change in variable, i.e.

$$\text{Growth Rate of } x = \frac{X_t - X_{t-1}}{X_{t-1} \times 100}$$

Where t indicates today and t-1 a year earlier.

Economic Growth and Equity Prices

Growth rates refer to the percentage change of a specific variable within a specific time period. Growth rates can be positive or negative, depending on whether the size of the variable is increasing or decreasing over time. Growth rates were first used by biologists studying population sizes, but they have since been brought into use in studying economic activity, corporate management, or investment returns.

For investors, growth rates typically represent the compounded annualised rate of growth of an investment, or a company's revenues, earnings, or dividends. Growth rates are also applied to more macro concepts, such as gross domestic product (GDP) and unemployment. Expected forward-looking or trailing growth rates are two common kinds of growth rates used for analysis.

- Growth rates are used to express the annual change in a variable as a percentage.
- A positive growth rate indicates a variable is increasing over time; a negative growth rate indicates that it is decreasing.
- Growth rates can be beneficial in assessing a company's performance and predicting future performance.
- Growth rates are computed by dividing the difference between the ending and starting values for the period being analysed and dividing that by the starting value.
- Time periods used for growth rates are most often annually, quarterly, monthly, and weekly.

The real economic growth rate, or real GDP growth rate, measures economic growth, as expressed by gross domestic product (GDP), from one period to another, adjusted for inflation or deflation. In other words, it reveals changes in the value of all goods and services produced by an economy—the economic output of a country—while accounting for price fluctuations.

$$\text{Real GDP growth rate} = \frac{\text{Most recent year's real GDP} - \text{the last year's real GDP}}{\text{The previous year's real GDP}}$$

Real economic growth can also be calculated by backing inflation out of nominal GDP. Nominal economic growth is inclusive of inflation, while real economic growth is not. This calculation is done by factoring in a GDP deflator. A GDP deflator is the quotient of nominal GDP divided by real GDP divided by 100, so this method is only useful in determining real GDP if the GDP deflator is already known.

$$\text{Real GDP} = (\text{Nominal GDP} \div \text{GDP Deflator}) \times 100$$

$$\text{Real GDP} = \frac{\text{Nominal GDP}}{\text{GDP Deflator}} \times 100$$

Limitations of Growth Rates

While growth rates are important for understanding how things change over time, they do come with some important limitations. First, the growth rate only considers the net change between two points in time, but it says nothing about the price movements or volatility that may have occurred in between. For instance, if some variable has a value of 10.00 today and 10.00 as well a year from now, the growth rate is zero. However, it could have fluctuated wildly—or not at all—during those 12 months. The annual growth rate, in this case, cannot tell us anything about that.

Growth rates also ignore the nominal amounts involved. For instance, Company A's earnings may grow from R100,000 per year to R150,000 per year, representing 50% growth, but only a R50,000 change. Much larger Company B's earnings may only grow at, say, 5% a year (10× less in terms of the growth rate) but amount to several millions of dollars in the company's coffers.

Finally, growth rates are hard to compare across industries or other unlike variables. A 5% rate of growth for a company may be relatively good or bad depending on if it is a growth-oriented tech startup vs. a large, incumbent consumer staples manufacturer. Likewise, a 4% decline in unemployment does not necessarily carry the same impact as a 4% increase in GDP.

1.5.6. INVESTORS AND ECONOMIC GROWTH

As indicated previously, growth in potential GDP represents the main driver of aggregate equity valuation. More generally, potential GDP also has implications for real interest rates. Positive growth in potential GDP indicates that future income will rise relative to current income. When consumers expect their incomes to rise, they increase current consumption and save less for future consumption (i.e., they are less likely to worry about funding their future consumption).

To encourage consumers to delay consumption (i.e., to encourage savings), investments would have to offer a higher real rate of return. Therefore, higher potential GDP growth implies higher real interest rates, and higher real asset returns in general.

In the short term, the relationship between actual GDP and potential GDP may provide insight to both equity and fixed-income investors as to the state of the economy. For example, since actual GDP in excess of potential GDP results in rising prices, the gap between the two can be used as a forecast of inflationary pressures—useful to all investors but of particular concern to fixed-income investors. Furthermore, central banks are likely to adopt monetary policies consistent with the gap between potential output and actual output. When actual GDP growth rate is higher (lower) than potential GDP growth rate, concerns about inflation increase (decrease) and the central bank is more likely to follow a restrictive (expansionary) monetary policy.

In addition to predicting monetary policy, the relationship between actual and potential GDP can also be useful in analysing fiscal policies. It is more likely for a government to run a fiscal deficit when actual GDP growth rate is lower than its potential growth rate.

Finally, because of the credit risk assumed by fixed-income investors, growth in GDP may be used to gauge credit risk of both corporate and government debt. A higher potential GDP growth rate reduces expected credit risk and generally increases the credit quality of all debt issues.

- Capital investment includes real estate, manufacturing plants, machinery, computers, vehicles, and production equipment.
- Economic growth can be measured by gross domestic product (GDP), the total monetary or market value of all the finished goods and services.
- In the U.S., economic growth is primarily driven by consumer spending and capital investment.

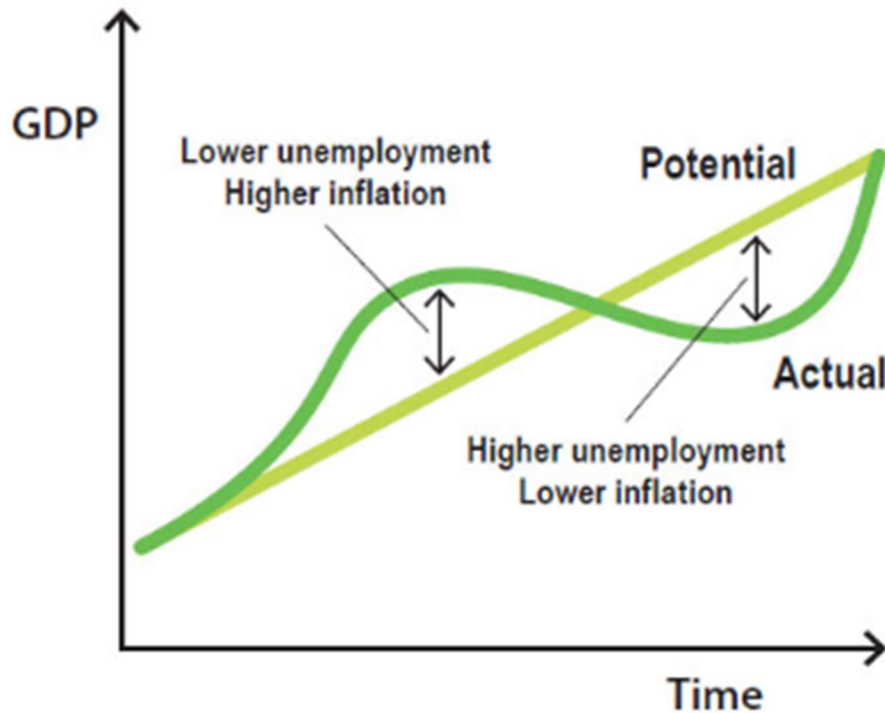
1.6. THE BUSINESS CYCLE

The business cycle refers to fluctuations in economic growth relative to growth in potential output. Monetary policy is one tool used to smooth fluctuations in the business cycle

In the upswing stage of the business cycle, there is usually strong growth in GDP and employment. As a result, the unemployment rate declines, and inflation starts to increase. The period of 2005–08 could be characterised as an upswing, as high commodity prices and mining investment contributed to strong economic growth. As the economy began to reach capacity constraints, wages growth and inflation began to increase. Around this time, the Reserve Bank increased the cash rate to put downward pressure on inflation.

In 2008, the global financial crisis put downward pressure on economic activity. Economic growth in Australia slowed, and we experienced other characteristics associated with a downturn. Growth in GDP and employment declined, and the unemployment rate increased. During this period, the cash rate was reduced to stimulate demand and support the economy.

The phases in the Business Cycle are illustrated below.



An expansion features growth in most sectors of the economy, with increasing employment, consumer spending, and business investment. As an expansion approaches its peak, the rates of increase in spending, investment, and employment slow but remains positive, while inflation accelerates.

A contraction or recession is associated with declines in most sectors, with inflation typically decreasing. When the contraction reaches a trough and the economy begins a new expansion or recovery, economic growth becomes positive again and inflation is typically moderate, but employment growth may not start to increase until the expansion has taken hold convincingly.

A common rule of thumb is to consider two consecutive quarters of growth in real GDP as the beginning of an expansion and two consecutive quarters of declining real GDP as indicating the beginning of a contraction.

Money and Business Cycles

Why do economies undergo such pronounced fluctuations?

- Evidence shows money plays an important role in generating business cycles, the upward and downward movement of aggregate output (the total production of goods and services), produced in the economy.
- When output is raising unemployment decreases, when output is falling, unemployment increases.
- Recessions are periods of declining aggregate output we see that the rate of money growth has declined before almost every recession indicating that changes in money might be the driving force behind business cycle fluctuations but not every decline in money growth is followed by a recession.



WATCH the Video:

Blockchain and Crypto

1.7. UNEMPLOYMENT

Unemployment is a term referring to individuals who are employable and actively seeking a job but are unable to find a job. Included in this group are those people in the workforce who are working but do not have an appropriate job. Usually measured by the unemployment rate, which is dividing the number of unemployed people by the total number of people in the workforce.

- Unemployment occurs when workers who want to work are unable to find jobs.
- High rates of unemployment signal economic distress while extremely low rates of unemployment may signal an overheated economy.
- Unemployment data is collected and published by government agencies in a variety of ways.
- Many governments offer unemployed individuals a small amount of income through unemployment insurance, as long as they meet certain requirements.

The **unemployment rate** is the percentage of people in the labour force who are unemployed. The labour force includes all people who are either *employed or unemployed*. People who choose not to be in the labour force are said to be voluntarily unemployed and are not included in the calculation of the unemployment rate.

A person who is employed part time but would prefer to work full time or is employed at a low-paying job despite being qualified for a significantly higher-paying one is said to be underemployed. Identification of the number of underemployed is somewhat subjective and not easily discernible from employment statistics.

Unemployment can be divided into three categories:

- **Frictional unemployment** results from the time lag necessary to match employees who seek work with employers needing their skills. Frictional unemployment is always with us as employers expand or contract their businesses and workers move, are fired, or quit to seek other opportunities.
- **Structural unemployment** is caused by long-run changes in the economy that eliminate some jobs while generating others for which unemployed workers are not qualified. Structural unemployment differs from frictional unemployment in that the unemployed workers do not currently have the skills needed to perform the jobs that are available.
- **Cyclical unemployment** is caused by changes in the general level of economic activity. Cyclical unemployment is positive when the economy is operating at less than full capacity and can be negative when an expansion leads to employment temporarily over the full employment level.

1.8. INFLATION

Inflation is a persistent increase in the price level over time. If the price level increases in a single jump but does not continue rising, the economy is not experiencing inflation. An increase in the price of a single good, or in relative prices of some goods, is not inflation. If inflation is present, the prices of almost all goods and services are increasing.

Inflation erodes the purchasing power of a currency. Inflation favours borrowers at the expense of lenders because when the borrower returns the principal to the lender, it is worth less in terms of goods and services (in real terms) than it was worth when it was borrowed. Inflation that accelerates out of control is referred to as hyperinflation, which can destroy a country's monetary system and bring about social and political upheavals.

The inflation rate is the percentage increase in the price level, typically compared to the prior year. Analysts can use the inflation rate as a business cycle indicator and to anticipate changes in central bank monetary policy. As we will see in the topic review on fiscal and monetary policy, an objective of central banks is to keep inflation within some target range. Disinflation refers to an inflation rate that is decreasing over time but remains greater than zero.

A persistently decreasing price level (i.e., a negative inflation rate) is called deflation. Deflation is commonly associated with deep recessions. When most prices are decreasing, consumers delay purchases because they believe they can buy the same goods more cheaply in the future. For firms, deflation results in decreasing revenue and increasing real fixed costs.

1.8.1. THE CONSUMER PRICE INDEX

To calculate a rate of inflation, we need to use a price index as a proxy for the price level. A price index measures the average price for a defined basket of goods and services.

The consumer price index (CPI) is the best-known indicator of SA inflation. Many countries use indexes similar to the CPI.

The CPI basket represents the purchasing patterns of a typical urban household.

To calculate the CPI, the Statistics Department compares the cost of the CPI basket today with the cost of the basket in an earlier base period.

$$\text{CPI} = \frac{\text{Cost of basket at current prices}}{\text{Cost of basket at base period prices}} = 100$$

For both consumer and producer prices, analysts and policymakers often distinguish between headline inflation and core inflation. Headline inflation refers to price indexes for all goods. Core inflation refers to price indexes that exclude food and energy. Food and energy prices are typically more volatile than those of most other goods. Thus, core inflation can sometimes be a more useful measure of the underlying trend in prices.

1.8.2. INVESTMENT CHARACTERISTICS

Inflation can be classified into three types: demand-pull inflation, cost-push inflation, and built-in inflation.

Cost-Push Inflation

- Inflation can result from an initial decrease in aggregate supply caused by an increase in the real price of an important factor of production, such as wages or energy.
- Cost-Push Inflation illustrates the effect on output and the price level of a decrease in aggregate supply.

Demand-Pull Inflation

- Demand-pull inflation can result from an increase in the money supply, increased government spending, or any other change that increases aggregate demand.
- Demand-Pull Inflation shows the effect on the price level when the economy is at full employment and aggregate demand increases.

Build-In Inflation

- Built-in inflation is related to adaptive expectations or the idea that people expect current inflation rates to continue in the future.
- As the price of goods and services rises, people may expect a continuous rise in the future at a similar rate.

Money and Inflation

The average price of goods and services in an economy is called the aggregate price level, or simply the price level.

Inflation is a continual increase in the price level and effect individuals, businesses and government.

What explains inflation?

- Data seems to indicate that a continuing increase in the money supply may be an important factor of increasing inflation.
- Evidence has found that the countries with the highest average inflation rate also have the highest interest rates

Effects of Inflation

Changes in price levels can affect economic growth because consumers and businesses may change the timing of their purchases, the amount of their spending and their saving and borrowing decisions based on expected changes in prices. The value of investments may also be affected by changing price levels.

Effect on consumers

- Consumers will spend rather than save
- Borrowers benefit because they repay loans with money that is worth less
- Wages may increase by less than inflation and thus consumers will have less money to spend

Effect on businesses

- Budgeting becomes difficult because of the uncertainty created by rising prices and costs
- Access to capital for firms is reduced because consumers spend rather than invest
- Profits may decline because of rising costs

Effect on investments

- The value of bonds will decline due to increases in interest rates as inflation increases ("inflation is the enemy of bonds")
- There will be a distribution of cash flows from bondholders to borrowers
- Shares may be protected against inflation. This will be the case if a company is able to pass price increases to consumers
- Commodities offer protection against inflation as they gain value when inflation goes up

1.8.3. MONETARY POLICY

Monetary policy refers to the central bank's actions that affect the quantity of money and credit in an economy in order to influence economic activity.

Monetary policy is said to be expansionary (or accommodative or easy) when the central bank increases the quantity of money and credit in an economy.

- Conversely, when the central bank is reducing the quantity of money and credit in an economy, the monetary policy is said to be contractionary (or restrictive or tight).
- Both monetary and fiscal policies are used by policymakers with the goals of maintaining stable prices and producing positive economic growth.

The Reserve Bank has three tools to influence money supply, and they are:

- **Open market operations:** The Reserve Bank increases money supply by buying treasury bills and reduces money supply by selling treasury bills. When they do the latter, they are reducing liquidity in the system, and when they do the former, they are injecting cash, so increasing the supply of money. A decrease in the supply of money is called contractionary monetary policy. An increase in the supply of money is called an expansionary monetary policy.
- **Control bank lending rates:** The Reserve Bank can also increase money supply by decreasing the interest rates it charges on loans to banks and reduce money supply by raising such interest rates. By, for example, decreasing interest rates, the Reserve Bank is encouraging commercial banks to lend more (as the interest rate they will charge the person they lend money to will also fall). Such a type of stimulus is also seen as expansionary.
- **Reserve requirement:** The reserve requirement is the percentage of customer deposits that banks are required to keep. The Reserve Bank will increase the reserve requirement to reduce money supply and reduce the reserve requirement to increase money supply. An increase in the reserve requirement naturally means that commercial banks have to hold more cash on hand so they will have less available to loan out.

Summary

- Monetary policy is a set of actions to control a nation's overall money supply and achieve economic growth.
- Monetary policy strategies include revising interest rates and changing bank reserve requirements.
- Monetary policy is commonly classified as either expansionary or contractionary.
- The Federal Reserve commonly uses three strategies for monetary policy including reserve requirements, the discount rate, and open market operations.

Money and Monetary Policy

Money is defined as anything that is generally accepted in payment for goods and services (in terms of its function as a medium of exchange).

In this section, the term money generally refers to the money supply.

Importance of Money:

- Money is linked to changes in economic variables and is important to the health of the economy.
- Money plays an important role in generating business cycles: empirical data indicates that, in the USA, the rate of growth in money supply has declined before every recession; however, not every decline in money growth is followed by a recession.
- Inflation is believed to be caused by continuing increases in money supply.
- Money plays an important role in interest rate fluctuations.

Limitations of Monetary Policy

The monetary policy can experience the following limitations on improving the economy of a country:

- **Time lags:** The effects of monetary policy can take months or years to materialise.
- **Technical limitations:** Interest rates can only be lowered to zero, which limits the effectiveness of monetary policy when rates are already low.
- **General impact:** Monetary policy tools affect the entire economy and can't be directed to specific problems or regions.
- **Risk of hyperinflation:** Setting interest rates too low can lead to over-borrowing and a speculative bubble.
- **Effectiveness depends on economic conditions:** The effectiveness of monetary policy depends on the state of the economy, including inflation expectations and financial markets.
- **Ineffective against other economic shocks:** Monetary policy may not be effective against other economic shocks, such as supply-side shocks.
- **Difficulty reversing deflation:** Central banks have limited ability to stimulate the economy once the nominal policy rate reaches zero.
- **Expectations gap:** There can be a gap between what monetary policy can deliver and what it's expected to deliver.

Monetary Policy Impact

If money neutrality holds, changes in monetary policy and the policy rate will have no effect on real output. In the short run, however, changes in monetary policy can affect real economic growth as well as interest rates, inflation, and foreign exchange rates. The effects of a change to a more expansionary monetary policy may include any or all of the following:

- The central bank buys securities, which increases bank reserves.
- The interbank lending rate decreases as banks are more willing to lend each other reserves.
- Other short-term rates decrease as the increase in the supply of loanable funds decreases the equilibrium rate for loans.
- Longer-term interest rates also decrease.
- The decrease in real interest rates causes the currency to depreciate in the foreign exchange market.
- The decrease in long-term interest rates increases business investment in plant and equipment.
- Lower interest rates cause consumers to increase their purchases of houses, autos, and durable goods.
- Depreciation of the currency increases foreign demand for domestic goods.
- These increases in consumption, investment, and net exports all increase aggregate demand.
- The increase in aggregate demand increases inflation, employment, and real GDP.

The transmission mechanism for a decrease in interbank lending rates affects four things simultaneously:

- Market rates decrease due to banks adjusting their lending rates for the short and long term.
- Asset prices increase because lower discount rates are used for computing present values.
- Firms and individuals raise their expectations for economic growth and profitability. They may also expect the central bank to follow up with further interest rate decreases.
- The domestic currency depreciates due to an outflow of foreign money as real interest rates decline.

Together, these four factors increase domestic demand as people consume more (they have less incentive to save given lower interest rates) and increase net external demand (exports minus imports) because depreciation of the domestic currency makes exports less expensive to foreigners and imports more expensive in the domestic economy. The increase in overall demand and import prices tends to increase aggregate demand and domestic inflation.

Advantages of Monetary Policy

- **Targeting an interest rate controls inflation:** A small amount of inflation is healthy for a growing economy as it encourages investment in the future and allows workers to expect higher wages. Inflation occurs when the general price levels of all goods and services in an economy increase. By raising the target interest rate, investment becomes more expensive and works to slow economic growth a bit.

- **Easy to implement:** Central banks can act quickly to use monetary policy tools. Often, just signaling their intentions to the market can yield results.
- **Central banks are independent and politically neutral:** Even if monetary policy action is unpopular, it can be undertaken before or during elections without the fear of political repercussions.
- **Weakening currency can boost exports:** Increasing the money supply or lowering interest rates tends to devalue the local currency. A weaker currency on world markets can serve to boost exports as these products are effectively less expensive for foreigners to purchase. The opposite effect would happen for companies that are mainly importers, hurting their bottom line.

Disadvantages of Monetary Policy

- **Effects have a time lag:** Even if implemented quickly, the macro effects of monetary policy generally occur after some time has passed. The effects on an economy may take months or even years to materialise. Some economists believe money is "merely a veil," and while serving to stimulate an economy in the short run, it has no long-term effects except for raising the general level of prices without boosting real economic output.
- **Technical limitations:** Interest rates can only be lowered nominally to 0%, which limits the bank's use of this policy tool when interest rates are already low. Keeping rates very low for prolonged periods of time can lead to a liquidity trap. This tends to make monetary policy tools more effective during economic expansions than recessions. Some European central banks have recently experimented with a negative interest rate policy (NIRP), but the results won't be known for some time to come.
- **Monetary tools are general and affect an entire country:** Monetary policy tools such as interest rate levels have an economy-wide impact and do not account for the fact some areas in the country might not need stimulus, while states with high unemployment might need the stimulus more. It is also general in the sense that monetary tools can't be directed to solve a specific problem or boost a specific industry or region.
- **Risk of hyperinflation:** When interest rates are set too low, over-borrowing at artificially cheap rates can occur. This can then cause a speculative bubble, whereby prices increase too quickly and to absurdly high levels. Adding more money to the economy can also run the risk of causing out-of-control inflation due to the premise of supply and demand: if more money is available in circulation, the value of each unit of money will decrease given an unchanged level of demand, making things priced in that money nominally more expensive.

The Bond Market and Interest Rates

The bond market is important because it enables corporations and governments to borrow money from investors to finance their activities. Interest rates are determined by the supply and demand for money as underpinned by the current economic conditions in a country.

An interest rate is the cost of borrowing, or the price paid for the rental of funds. Interest rates are important because:

- Bonds are subject to interest rate risk since rising rates will result in falling prices (and vice-versa).
- Interest rates respond to inflation: when prices in an economy rise, the central bank typically raises its target rate to cool down an overheating economy.

- International Monetary Fund. "Monetary Policy: Stabilising Prices and Output."
- Inflation also erodes the real value of a bond's face value, which is a particular concern for longer maturity debts.
- Because of these linkages, bond prices are quite sensitive to changes in inflation and inflation forecasts.

Inflation is a bond's worst enemy. Inflation erodes the purchasing power of a bond's future cash flows. Typically, bonds are fixed-rate investments. If inflation is increasing (or rising prices), the return on a bond is reduced in real terms, meaning adjusted for inflation. For example, if a bond pays a 4% yield and inflation is 3%, the bond's real rate of return is 1%.

1.8.4. FISCAL POLICY

Fiscal policy refers to a government's use of spending and taxation to influence economic activity. The budget is said to be balanced when tax revenues equal government expenditures. A budget surplus occurs when government tax revenues exceed expenditures, and a budget deficit occurs when government expenditures exceed tax revenues.

Both monetary and fiscal policies are used by policymakers with the goals of maintaining stable prices and producing positive economic growth. Fiscal policy can also be used as a tool for redistribution of income and wealth.

Increasing taxes will have the following impact on the economy:

- A decrease in the disposable income of consumers, since they will pay more tax, either through increased VAT or increased personal income tax (or both); This means that there will be a...
- A decrease in consumer spending, as consumers will have a lower disposable income. Lower consumer spending ultimately also leads to less production by firms, with the result that there is a decrease in economic growth

Should taxes be lowered, the opposite will occur. Raising taxes or reducing government spending is called contractionary fiscal policy, and reducing taxes or increasing government spending is called expansionary fiscal policy.

Increasing government spending will have the following effect on the economy:

- Incomes will increase
- Demands for goods and services will increase; production by firms will rise as a result; and
- The economy will grow

Limitations Of Fiscal Policy

Fiscal policy can experience the following limitations on improving the economy of a country:

- **Time lags:** It takes time for policy makers to recognise that a problem exists, for decisions to be made and implemented, and for actions to occur that will have an impact on the economy. By the time the actions take effect, the economy and economic conditions may have already changed.

- **Unexpected responses:** Consumers and companies may not respond as expected to changes in fiscal policy. For example, with a reduction in taxes, consumers and companies may choose to save rather than spend as would be expected by the policy change.
- **Unintended consequences:** Government spending may lead to economic growth in the short term but higher inflation rates in the long run.

Government is usually careful to try and come close to balancing its spending and its income. Too much spending causes a deficit which has to be financed somehow, and financing naturally comes with a cost: the government will have to pay interest on its debt. The higher the degree of debt in a country, the higher are its debt-servicing costs (interest payments), meaning that there is less money available for other needs the country might face (like improving education, building houses, hospitals etc.).

Both fiscal and monetary policy have economy-wide effects, and effects on the financial services industry. Consumer spending cuts across spending on houses, loans, food, cars etc., so changes in economic policy through changes in fiscal and monetary policy end up having real effects on households and companies. The financial sector is not only directly impacted by changes to taxes and interest rates, but also indirectly through changes in consumer spending also.

Advantages Of Fiscal Policy

- **Can direct spending to specific purposes:** Unlike monetary policy tools, which are general in nature, a government can direct spending toward specific projects, sectors, or regions to stimulate the economy where it is perceived to be needed most.
- **Can use taxation to discourage negative externalities:** Taxing polluters or those that overuse limited resources can help remove the negative effects they cause while generating government revenue.
- **Short time lag:** The effects of fiscal policy tools can be seen much quicker than the effects of monetary tools.

Disadvantages Of Fiscal Policy

- **May be politically motivated:** Raising taxes can be unpopular and politically dangerous to implement.
- **Tax incentives may be spent on imports:** The effect of fiscal stimulus is muted when the money put into the economy through tax savings or government spending is spent on imports, sending that money abroad instead of keeping it in the local economy.
- **Can create budget deficits:** A government budget deficit is when it spends more money annually than it takes in. If spending is high and taxes are low for too long, such a deficit can continue to widen to dangerous levels.